

$a, T: R^2 \rightarrow R^3, T(x, y) = (x - y, 2x + y, 0)$
 $T(x, y) = (ax + by, cx + dy, ex + fy)$
 $T(0, 0) = (0 - 0, 2 \cdot 0 + 0, 0)$
 $T(0, 0) = (0, 0, 0) \in \text{linear}$

$b, T: R^2 \rightarrow R^2, T(x, y) = (x + 2, y + 3)$
 $T(x, y) = (ax + by, cx + dy)$
 $T(0, 0) \neq (2, 3) \times \text{Not } \mathbb{R}\text{-linear}$

$c, R^2 \rightarrow R, T(x, y) = |x|$
 $T(-2, 10) = |-2| = 2$
 $|a+b| \neq |a| + |b|$
 $v = (1, 0) \quad k = -1$
 $T(kv) = -1 \times \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$
 $T(-1, 0) = |-1| = 1$
 $T(1, 0) = |1| = 1$
 $kT(v) = -1 \cdot 1 = -1 \quad 1 \neq -1 \text{ not linear}$

$d, T: M_{nn} \rightarrow M_{nn}, T\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = 3a - 4b + (c - d)$
 $0 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
 $= 3 \cdot 0 - 4 \cdot 0 + 0 - 0 = 0$
 $\in \text{Linear}$

$e, P_2 \rightarrow P_2, T(a_0 + a_1x + a_2x^2) = a_0 + a_1(x+1) + a_2(x+1)^2$
 $x^2 = (x+1)^2 = x^2 + 2x + 1$
 $T(0) = 0 + 0(x+1) + 0(x+1)^2 = 0$
 $T(0(x)) = 0(x+1) \in \text{linear}$

$f, T: P_2 \rightarrow P_2, T(a_0 + a_1x + a_2x^2) = (a_0 + a_1) + (a_1 + a_2)x + (a_2 + 1)x^2$
 $T(0) = (0+1) + (0+0)x + (0+1)x^2$
 $T(0) = 1 + 0x + 1x^2 = 1 + x^2$
 $\text{Not } \mathbb{C}\text{-linear}$

$g, T: R^2 \rightarrow R^2, T(x, y) = (x^2, y^2)$
 $T(x, y) = (ax + by, cx + dy)$
 $\text{Not } \mathbb{R}\text{-linear, form substitution}$

$h, B = \{v_1, v_2\} \subset R^2 \quad v_1 = (-2, 7) \quad v_2 = (1, 3) \quad R^2 \rightarrow R^2$
 $T(v_1) = (1, -2, 0) \quad T(v_2) = (0, -3, 5)$
 $T(x, y) = ? \quad \begin{cases} -2a + b = x \\ a + 3b = y \end{cases}$
 $-2a + b = x \quad a + 3b = y$
 $a = y - 3b$
 $-2(y - 3b) + b = x$
 $-2y + 6b + b = x$
 $-2y + 7b = x$
 $7b = x + 2y$
 $b = \frac{x + 2y}{7}$
 $a = y - 3 \cdot \frac{x + 2y}{7} = y - \frac{3x + 6y}{7} = \frac{7y - 3x - 6y}{7} = \frac{-3x + y}{7}$
 $b = \frac{x + 2y}{7} \quad a = \frac{-3x + y}{7}$

$T(x, y) = aT(v_1) + bT(v_2)$
 $T(x, y) = \left(\frac{-3x + y}{7}\right)(1, -2, 0) + \left(\frac{x + 2y}{7}\right)(0, -3, 5)$
 $= \left(\frac{-3x + y}{7}, \frac{6x - 2y}{7}, 0\right) + \left(0, \frac{-3x - 6y}{7}, \frac{5x + 10y}{7}\right)$
 $= \left(\frac{-3x + y}{7}, \frac{6x - 2y - 3x - 6y}{7}, \frac{5x + 10y}{7}\right)$
 $= \left(\frac{-3x + y}{7}, \frac{3x - 8y}{7}, \frac{5x + 10y}{7}\right)$
 $\frac{-3x + y}{7} + 0 = \frac{-3x + y}{7}$
 $\frac{6x - 2y}{7} + \frac{-3x - 6y}{7} = \frac{3x - 8y}{7}$
 $0 + \frac{5x + 10y}{7} = \frac{5x + 10y}{7}$

$i, T: R^2 \rightarrow R^2, T(x, y) = (x + 2, y + 3)$
 $T(x, y, z) = ?$
 $T(v_1) = (2, -1, 4)$
 $T(v_2) = (3, 0, 1)$
 $T(v_3) = (-1, 5, 1)$
 $a + b + c = x$
 $a + b = y$
 $a = z$
 $a + b = y \Leftrightarrow z + b = y \Leftrightarrow b = y - z$
 $T(x, y, z) = (-x + 4y - 2z, 5x - 2y + 3z)$

$h, T: R^2 \rightarrow R^2, T(x, y) = (y - x, 0)$
 $T(x, y) = (ax + by, cx + dy)$
 $v = 1, 2$
 $k = 3$
 $T(1, 2) = (2 - 1, 0)$
 $= (1, 0)$
 $T(kv) = T(3, 6) = (6 - 3, 0)$
 $= (3, 0)$
 $kT(v) = 3 \times (1, 0)$
 $= (3, 0)$
 $T(kv) = kT(v)$
 $\in \text{Linear}$

$i, T: R^3 \rightarrow R^4, T(x, y) = (y, x, y, x)$
 $T(x, y) = (ax + by, cx + dy, ex + fy, gx + hy)$
 $\in \text{Linear}$
 $v = (5, 3)$
 $T(5, 3) = (3, 5, 3, 5)$
 $u = (2, 1)$
 $T(u) = (1, 2, 1, 2)$
 $T(u) + T(v) = (3, 5, 3, 5) + (1, 2, 1, 2)$
 $= (4, 7, 4, 7)$
 $T(u+v) = T((5, 3) + (2, 1))$
 $T(7, 4) = (4, 7, 4, 7)$
 $T(u) + T(v) = T(u+v)$
 $\in \text{Linear}$

$j, M_{2 \times 2}, T(x, y) = \begin{pmatrix} 2x & 3y \\ -y & x+y \end{pmatrix}$
 $T(u) = \begin{pmatrix} 2 \cdot 5 & 3 \cdot 3 \\ -3 & 5+3 \end{pmatrix} = \begin{pmatrix} 10 & 9 \\ -3 & 8 \end{pmatrix}$
 $T(v) = \begin{pmatrix} 2 \cdot 1 & 3 \cdot 2 \\ -2 & 1+2 \end{pmatrix} = \begin{pmatrix} 2 & 6 \\ -2 & 3 \end{pmatrix}$
 $T(u+v) = T\left(\begin{pmatrix} 6 & 5 \\ -1 & 5 \end{pmatrix}\right) = \begin{pmatrix} 12 & 15 \\ -5 & 10 \end{pmatrix}$
 $T(u) + T(v) = \begin{pmatrix} 10 & 9 \\ -3 & 8 \end{pmatrix} + \begin{pmatrix} 2 & 6 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} 12 & 15 \\ -5 & 11 \end{pmatrix}$
 $\neq T(u+v)$
 $\notin \text{Linear}$

$k, T: R^3 \rightarrow R^2, T(x, y, z) = \begin{bmatrix} 2 & 1 & 3 \\ -1 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$
 $T(x, y, z) = \begin{bmatrix} 2x + y + 3z \\ -x - 2z \end{bmatrix}$
 $T(x, y, z) = (2x + y + 3z, -x - 2z)$
 $T(x + y + z) = T(x) + T(y) + T(z)$

$v = (2, 1)$
 $u = (3, 2)$
 $w = (2, 5)$
 $T(v) = (2 \cdot 2 + 1 \cdot 1, -2 - 2 \cdot 1) = (5, -4)$
 $T(u) = (2 \cdot 3 + 2 \cdot 2, -3 - 2 \cdot 2) = (10, -7)$
 $T(w) = (2 \cdot 2 + 5 \cdot 1, -2 - 2 \cdot 5) = (9, -12)$
 $T(v) + T(u) = (5, -4) + (10, -7) = (15, -11)$
 $T(v+u) = T((2, 1) + (3, 2)) = T(5, 3) = (5 \cdot 2 + 3 \cdot 1, -5 - 2 \cdot 3) = (13, -11)$
 $\neq T(v) + T(u)$
 $\notin \text{Linear}$

$l, T: R^3 \rightarrow R^2, T(x, y, z) = \begin{bmatrix} 2 & 1 & 3 \\ -1 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$
 $T(x, y, z) = (2x + y + 3z, -x - 2z)$
 $T(x + y + z) = T(x) + T(y) + T(z)$

$v = (2, 1)$
 $u = (3, 2)$
 $w = (2, 5)$
 $T(v) = (2 \cdot 2 + 1 \cdot 1, -2 - 2 \cdot 1) = (5, -4)$
 $T(u) = (2 \cdot 3 + 2 \cdot 2, -3 - 2 \cdot 2) = (10, -7)$
 $T(w) = (2 \cdot 2 + 5 \cdot 1, -2 - 2 \cdot 5) = (9, -12)$
 $T(v) + T(u) = (5, -4) + (10, -7) = (15, -11)$
 $T(v+u) = T((2, 1) + (3, 2)) = T(5, 3) = (5 \cdot 2 + 3 \cdot 1, -5 - 2 \cdot 3) = (13, -11)$
 $\neq T(v) + T(u)$
 $\notin \text{Linear}$

$m, T: R^3 \rightarrow R^2, T(x, y, z) = \begin{bmatrix} 2 & 1 & 3 \\ -1 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$
 $T(x, y, z) = (2x + y + 3z, -x - 2z)$
 $T(x + y + z) = T(x) + T(y) + T(z)$

$n, T: R^3 \rightarrow R^2, T(x, y, z) = \begin{bmatrix} 2 & 1 & 3 \\ -1 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$
 $T(x, y, z) = (2x + y + 3z, -x - 2z)$
 $T(x + y + z) = T(x) + T(y) + T(z)$

$1, T: R^2 \rightarrow R^2, T(x, y) = (y - x, 0)$
 $T(1, 1) = (1 - 1, 0) = (0, 0)$
 $T(-1, 1) = (1 - (-1), 0) = (2, 0)$
 $T(x, y) = ?$

$(x, y) = a(1, 1) + b(-1, 1)$
 $\begin{cases} a - b = x \\ -a + 2b = y \end{cases}$
 $a - b = x$
 $\Leftrightarrow b = a - x$
 $-a + 2(a - x) = y$
 $-a + 2a - 2x = y$
 $a - 2x = y$
 $a = y + 2x$

$b = a - x = (y + 2x) - x = x + y$
 $a = y + 2x$
 $b = x + y$
 $T(x, y) = T[a(1, 1) + b(-1, 1)] = aT(1, 1) + bT(-1, 1)$
 $T(x, y) = (y + 2x) \cdot (2, 0) + (x + y) \cdot (0, 0)$
 $= (2y + 4x, 0)$
 $T(x, y) = 2y + 4x$

$2, T: R^2 \rightarrow R^2, T(x, y) = (y - x, 0)$
 $T(1, 0) = (0 - 1, 0) = (-1, 0)$
 $T(0, 1) = (1 - 0, 0) = (1, 0)$
 $T(x, y) = ?$
 $\begin{cases} a = x \\ b = y \end{cases}$
 $T(x, y) = T[a(1, 0) + b(0, 1)]$

$T(x, y) = x(1, 0) + y(0, 1)$
 $= (x, 0) + (0, y)$
 $= (x, y)$
 $T(x, y) = (x, y)$

$3, T: R^2 \rightarrow R^2, T(x, y) = (y - x, 0)$
 $T(1, 0) = (0 - 1, 0) = (-1, 0)$
 $T(0, 1) = (1 - 0, 0) = (1, 0)$
 $T(x, y) = ?$
 $\begin{cases} a = x \\ b = y \end{cases}$
 $T(x, y) = T[a(1, 0) + b(0, 1)]$

$4, B = \{v_1, v_2\} \subset R^2 \quad v_1 = (1, 1) \quad v_2 = (1, -2)$
 $T: R^2 \rightarrow R^2 \quad T(v_1) = (1, -2) \quad T(v_2) = (4, -1)$
 $\begin{cases} a + b = x \\ a = y \end{cases}$
 $a = y$
 $y + b = x$
 $b = x - y$
 $a = y$

$T(x, y) = T[a(v_1) + b(v_2)] = aT(v_1) + bT(v_2)$
 $a \cdot (1, -2) = y(1, -2)$
 $= (y, -2y)$
 $b \cdot (4, -1) = (x - y)(4, -1)$
 $= (4x - 4y, -x + y)$
 $T(x, y) = (y + 4x - 4y, -2y - x + y) = (4x - 3y, -x - y)$

$T(x, y) = (4x - 3y, -x - y)$
 $T(x, y) = (4x - 3y, -x - y)$

$5, T: R^3 \rightarrow R^2, T(x, y, z) = \begin{bmatrix} 2 & 1 & 3 \\ -1 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$
 $T(x, y, z) = (2x + y + 3z, -x - 2z)$
 $T(x + y + z) = T(x) + T(y) + T(z)$